CHAPTER

Introduction to Databases

Chapter Objectives

In this chapter you will learn:

- The importance of database systems.
- Some common uses of database systems.
- The characteristics of file-based systems.
- The problems with the file-based approach.
- The meaning of the term "database."
- The meaning of the term "database management system" (DBMS).
- The typical functions of a DBMS.
- The major components of the DBMS environment.
- The personnel involved in the DBMS environment.
- The history of the development of DBMSs.
- The advantages and disadvantages of DBMSs.

While barely 50 years old, database research has had a profound impact on the economy and society, creating an industry sector valued at between US$35-US$50 billion annually. The database system, the subject of this book, is arguably the most important development in the field of software engineering, and the database is now the underlying framework of the information system, fundamentally changing the way that many organizations operate. Indeed the importance of the database system has increased in recent years with significant developments in hardware capability, hardware capacity, and communications, including the emergence of the Internet, electronic commerce, business intelligence, mobile communications, and grid computing.

Database technology has been an exciting area to work in and, since its emergence, has been the catalyst for many important developments in software engineering. Database research is not over and there are still many problems that need

to be addressed. Moreover, as the applications of database systems become even more complex we will have to rethink many of the algorithms currently being used, such as the algorithms for file storage, file access, and query optimization. These original algorithms have made significant contributions in software engineering and, without doubt, the development of new algorithms will have similar effects. In this first chapter, we introduce the database system.

"As a result of its importance, many computing and business courses cover the study of database systems. In this book, we explore a range of issues associated with the implementation and applications of database systems. We also focus on how to design a database and present a methodology that should help you design both simply and complex databases.

Structure of this Chapter In Section 1.1, we examine some uses of database systems that we find in everyday life but are not necessarily aware of. In Sections 1.2 and 1.3, we compare the early file-based approach to computerizing the manual file system with the modern, and more usable, database approach. In Section 1.4, we discuss the four types of role that people perform in the database environment, namely: data and database administrators, database designers, application developers, and end-users. In Section 1.5, we provide a brief history of database systems, and follow that in Section 1.6 with a discussion of the advantages and disadvantages of database systems.

Throughout this book, we illustrate concepts using a case study based on a fictitious property management company called *DreamHome*. We provide a detailed description of this case study in Section 11.4 and Appendix A. In Appendix B, we present further case studies that are intended to provide additional realistic projects for the reader. There will be exercises based on these case studies at the end of many chapters.

1.1 Introduction

The database is now such an integral part of our day-to-day life that often we are not aware that we are using one. To start our discussion of databases, in this section we examine some applications of database systems. For the purposes of this discussion, we consider a **database** to be a collection of related data and a **database management system (DBMS)** to be the software that manages and controls access to the database. A **database application** is simply a program that interacts with the database at some point in its execution. We also use the more inclusive term **database system** as a collection of application programs that interact with the database along with the DBMS and the database itself. We provide more accurate definitions in Section 1.3.

Purchases from the supermarket

When you purchase goods from your local supermarket, it is likely that a database is accessed. The checkout assistant uses a bar code reader to scan each of your purchases. This reader is linked to a database application that uses the bar code to find out the price of the item from a product database. The application then reduces the number of such items in stock and displays the price on the cash register. If the reorder level falls below a specified threshold, the database system may automatically place an order to obtain more of that item. If a customer telephones the supermarket, an assistant can check whether an item is in stock by running an application program that determines availability from the database.

Purchases using your credit card

When you purchase goods using your credit card, the assistant normally checks whether you have sufficient credit left to make the purchase. This check may be carried out by telephone or automatically by a card reader linked to a computer system. In either case, there is a database somewhere that contains information about the purchases that you have made using your credit card. To check your credit, there is a database application that uses your credit card number to check that the price of the goods you wish to buy, together with the sum of the purchases that you have already made this month, is within your credit limit. When the purchase is confirmed, the details of the purchase are added to this database. The database application also accesses the database to confirm that the credit card is not on the list of stolen or lost cards before authorizing the purchase. There are other database applications to send out monthly statements to each cardholder and to credit accounts when payment is received.

Booking a vacation with a travel agent

When you make inquiries about a vacation, your travel agent may access several databases containing vacation and flight details. When you book your vacation, the database system has to make all the necessary booking arrangements. In this case, the system has to ensure that two different agents do not book the same vacation or overbook the seats on the flight. For example, if there is only one seat left on the flight from New York to London and two agents try to reserve the last seat at the same time, the system has to recognize this situation, allow one booking to proceed, and inform the other agent that there are now no seats available. The travel agent may have another, usually separate, database for invoicing.

Using the local library

Your local library probably has a database containing details of the books in the library, details of the readers, reservations, and so on. There will be a computerized index that allows readers to find a book based on its title, authors, or subject area. The database system handles reservations to allow a reader to reserve a book and to be informed by mail or email when the book is available. The system also sends reminders to borrowers who have failed to return books by the due date. Typically, the system will have a bar code reader, similar to that used by the

supermarket described earlier, that is used to keep track of books coming in and going out of the library.

Taking out insurance

Whenever you wish to take out insurance—for example personal insurance, property insurance, or auto insurance, your agent may access several databases containing figures for various insurance organizations. The personal details that you supply, such as name, address, age, and whether you drink or smoke, are used by the database system to determine the cost of the insurance. An insurance agent can search several databases to find the organization that gives you the best deal.

Renting a DVD

When you wish to rent a DVD from a DVD rental company, you will probably find that the company maintains a database consisting of the DVD titles that it stocks, details on the copies it has for each title, whether the copy is available for rent or is currently on loan, details of its members (the renters), and which DVDs they are currently renting and date they are returned. The database may even store more detailed information on each DVD, such as its director and its actors. The company can use this information to monitor stock usage and predict future buying trends based on historic rental data.

Using the Internet

Many of the sites on the Internet are driven by database applications. For example, you may visit an online bookstore that allows you to browse and buy books, such as Amazon.com. The bookstore allows you to browse books in different categories, such as computing or management, or by author name. In either case, there is a database on the organization's Web server that consists of book details, availability, shipping information, stock levels, and order history. Book details include book titles, ISBNs, authors, prices, sales histories, publishers, reviews, and detailed descriptions. The database allows books to be cross-referenced: for example, a book may be listed under several categories, such as computing, programming languages, bestsellers, and recommended titles. The cross-referencing also allows Amazon to give you information on other books that are typically ordered along with the title you are interested in.

As with an earlier example, you can provide your credit card details to purchase one or more books online. Amazon.com personalizes its service for customers who return to its site by keeping a record of all previous transactions, including items purchased, shipping, and credit card details. When you return to the site, you might be greeted by name and presented with a list of recommended titles based on previous purchases.

Studying at College

If you are at college, there will be a database system containing information about yourself, your major and minor fields, the courses you are enrolled in, details about your financial aid, the classes you have taken in previous years or are taking

this year, and details of all your examination results. There may also be a database containing details relating to the next year's admissions and a database containing details of the staff working at the university, giving personal details and salary-related details for the payroll office.

These are only a few of the applications for database systems, and you will no doubt know of plenty of others. Though we now take such applications for granted, the database system is a highly complex technology that has been developed over the past 40 years. In the next section, we discuss the precursor to the database system: the file-based system.

1.2 Traditional File-Based Systems

It is almost a tradition that comprehensive database books introduce the database system with a review of its predecessor, the file-based system. We will not depart from this tradition. Although the file-based approach is largely obsolete, there are good reasons for studying it:

- Understanding the problems inherent in file-based systems may prevent us from repeating these problems in database systems. In other words, we should learn from our earlier mistakes. Actually, using the word "mistakes" is derogatory and does not give any cognizance to the work that served a useful purpose for many years. However, we have learned from this work that there are better ways to handle data.
- If you wish to convert a file-based system to a database system, understanding how the file system works will be extremely useful, if not essential.

1.2.1 File-Based Approach

File-based system A collection of application programs that perform services for the end-users, such as the production of reports. Each program defines and manages its own data.

File-based systems were an early attempt to computerize the manual filing system that we are all familiar with. For example, an organization might have physical files set up to hold all external and internal correspondence relating to a project, product, task, client, or employee. Typically, there are many such files, and for safety they are labeled and stored in one or more cabinets. For security, the cabinets may have locks or may be located in secure areas of the building. In our own home, we probably have some sort of filing system that contains receipts, warranties, invoices, bank statements, and so on. When we need to look something up, we go to the filing system and search through the system, starting at the first entry, until we find what we want. Alternatively, we might have an indexing system that helps locate what we want more quickly. For example, we might have divisions in the filing system or separate folders for different types of items that are in some way *logically related*.

The manual filing system works well as long as the number of items to be stored is small. It even works adequately when there are large numbers of items and we only have to store and retrieve them. However, the manual filing system breaks down when we have to cross-reference or process the information in the files. For

example, a typical real estate agent's office might have a separate file for each property for sale or rent, each potential buyer and renter, and each member of staff. Consider the effort that would be required to answer the following questions:

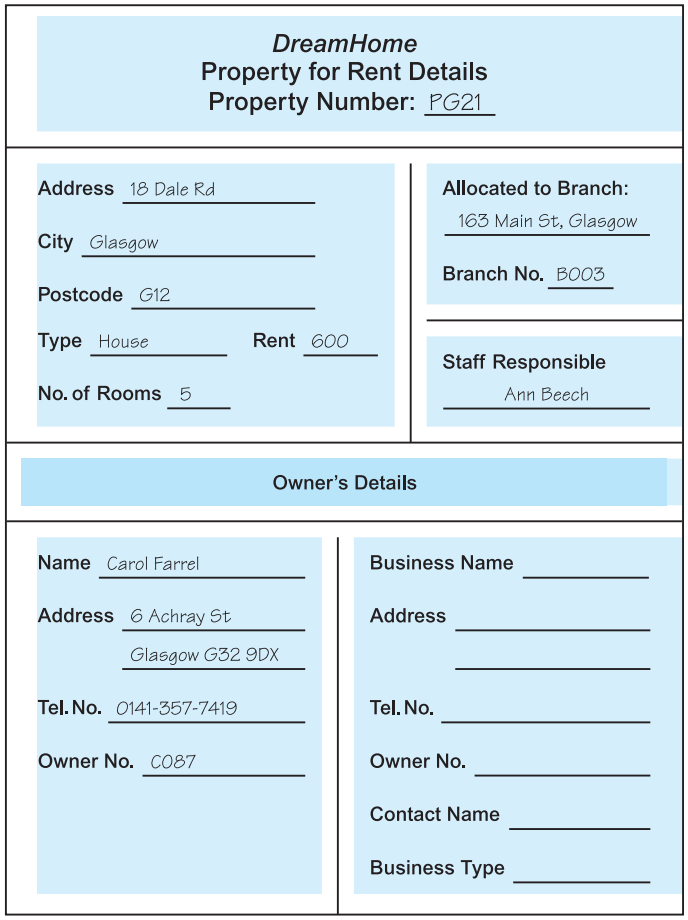
- What three-bedroom properties do you have for sale with an acre of land and a garage?
- What apartments do you have for rent within three miles of downtown?
- What is the average rent for a two-bedroom apartment?
- What is the annual total for staff salaries?
- How does last month's net income compare with the projected figure for this month?
- What is the expected monthly net income for the next financial year?

Increasingly nowadays, clients, senior managers, and staff want more and more information. In some business sectors, there is a legal requirement to produce detailed monthly, quarterly, and annual reports. Clearly, the manual system is inadequate for this type of work. The file-based system was developed in response to the needs of industry for more efficient data access. However, rather than establish a centralized store for the organization's operational data, a decentralized approach was taken, where each department, with the assistance of **Data Processing (DP)** staff, stored and controlled its own data. To understand what this means, consider the *DreamHome* example.

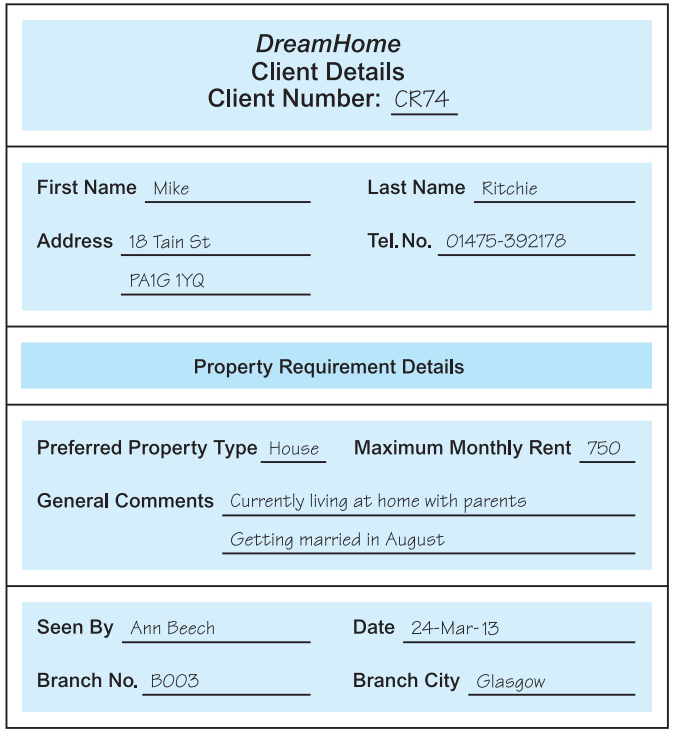
The Sales Department is responsible for the selling and renting of properties. For example, whenever a client who wishes to offer his or her property as a rental approaches the Sales Department, a form similar to the one shown in Figure 1.1(a) is completed. The completed form contains details on the property, such as address, number of rooms, and the owner's contact information. The Sales Department also handles inquiries from clients, and a form similar to the one shown in Figure l.l(b) is completed for each one. With the assistance of the DP Department, the Sales Department creates an information system to handle the renting of property. The system consists of three files containing property, owner, and client details, as illustrated in Figure 1.2. For simplicity, we omit details relating to members of staff, branch offices, and business owners.

The Contracts Department is responsible for handling the lease agreements associated with properties for rent. Whenever a client agrees to rent a property, a form with the client and property details is filled in by one of the sales staff, as shown in Figure 1.3. This form is passed to the Contracts Department, which allocates a lease number and completes the payment and rental period details. Again, with the assistance of the DP Department, the Contracts Department creates an information system to handle lease agreements. The system consists of three files that store lease, property, and client details, and that contain similar data to that held by the Sales Department, as illustrated in Figure 1.4

The process is illustrated in Figure 1.5. It shows each department accessing their own files through application programs written especially for them. Each set of departmental application programs handles data entry, file maintenance, and the generation of a fixed set of specific reports. More important, the physical structure and storage of the data files and records are defined in the application code.



(a)



(b)

Figure 1.1 Sales Department forms; (a) Property for Rent Details form; (b) Client Details form.

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead Rd	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

Client

clientNo	fName	lName	address	telNo	prefType	maxRent
CR76	John	Kay	56 High St, London SW1 4EH	0207-774-5632	Flat	425
CR56	Aline	Stewart	64 Fern Dr, Glasgow G42 0BL	0141-848-1825	Flat	350
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178	House	750
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720	Flat	600

Figure 1.2 The PropertyForRent, PrivateOwner, and Client files used by Sales.

Figure 1.3 Lease Details form used by Contracts Department.

DreamHome
Lease Details
Lease Number: 10012

Client No. CR74
Full Name Mike Ritchie
Address (previous) 18 Tain St
PA1G 1YQ
Tel. No. 01475-392178

Property No. PG21
Address 18 Dale Rd
Glasgow G12

Payment Details

Monthly Rent 600
Payment Method Cheque
Deposit 1200 Paid (Y or N) Y

Rent Start Date 1-Jul-13
Rent Finish Date 30-Jun-14
Duration 1 Year

Lease

leaseNo	propertyNo	clientNo	rent	payment Method	deposit	paid	rentStart	rentFinish	duration
10024	PA14	CR62	650	Visa	1300	Y	1-Jun-13	31-May-14	12
10075	PL94	CR76	400	Cash	800	N	1-Aug-13	31-Jan-14	6
10012	PG21	CR74	600	Cheque	1200	Y	1-Jul-13	30-Jun-14	12

PropertyForRent

propertyNo	street	city	postcode	rent
PA14	16 Holhead	Aberdeen	AB7 5SU	650
PL94	6 Argyll St	London	NW2	400
PG21	18 Dale Rd	Glasgow	G12	600

Client

clientNo	fName	lName	address	telNo
CR76	John	Kay	56 High St, London SW1 4EH	0171-774-5632
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720

Figure 1.4 The Lease, PropertyForRent, and Client files used by the Contracts Department.

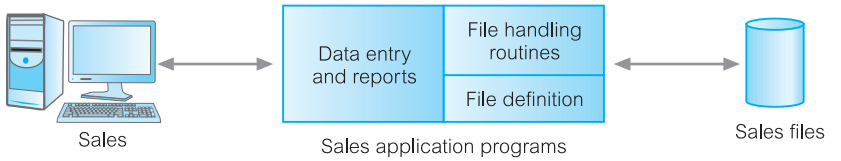


Figure 1.5 File-based processing.

Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

We can find similar examples in other departments. For example, the Payroll Department stores details relating to each member of staff's salary:

StaffSalary(staffNo, fName, lName, sex, salary, branchNo)

The Human Resources (HR) Department also stores staff details:

Staff(staffNo, fName, lName, position, sex, dateOfBirth, salary, branchNo)

It can be seen quite clearly that there is a significant amount of duplication of data in these departments, and this is generally true of file-based systems. Before we discuss the limitations of this approach, it may be useful to understand the terminology used in file-based systems. A file is simply a collection of **records**, which contains **logically related data**. For example, the PropertyForRent file in Figure 1.2 contains six records, one for each property. Each record contains a logically connected set of one or more **fields**, where each field represents some characteristic of the real-world object that is being modeled. In Figure 1.2, the fields of the PropertyForRent file represent characteristics of properties, such as address, property type, and number of rooms.

1.2.2 Limitations of the File-Based Approach

This brief description of traditional file-based systems should be sufficient to discuss the limitations of this approach. We list five problems in Table 1.1.

Separation and isolation of data

When data is isolated in separate files, it is more difficult to access data that should be available. For example, if we want to produce a list of all houses that match the requirements of clients, we first need to create a temporary file of those clients who have "house" as the preferred type. We then search the PropertyForRent file for those properties where the property type is "house" and the rent is less than the client's maximum rent. With file systems, such processing is difficult. The application developer must synchronize the processing of two files to ensure that the correct data is extracted. This difficulty is compounded if we require data from more than two files.

Duplication of data

Owing to the decentralized approach taken by each department, the file-based approach encouraged, if not necessitated, the uncontrolled duplication of data. For example, in Figure 1.5 we can clearly see that there is duplication of both property and client details in the Sales and Contracts Departments. Uncontrolled duplication of data is undesirable for several reasons, including:

- Duplication is wasteful. It costs time and money to enter the data more than once.
- It takes up additional storage space, again with associated costs. Often, the duplication of data can be avoided by sharing data files.

TABLE 1.1 Limitations of file-based systems.

Separation and isolation of data
Duplication of data
Data dependence
Incompatible file formats
Fixed queries/proliferation of application programs

- Perhaps more importantly, duplication can lead to loss of data integrity; in other words, the data is no longer consistent. For example, consider the duplication of data between the Payroll and HR Departments described previously. If a member of staff moves and the change of address is communicated only to HR and not to Payroll, the person's paycheck will be sent to the wrong address. A more serious problem occurs if an employee is promoted with an associated increase in salary. Again, let's assume that the change is announced to HR, but the change does not filter through to Payroll. Now, the employee is receiving the wrong salary. When this error is detected, it will take time and effort to resolve. Both these examples illustrate inconsistencies that may result from the duplication of data. As there is no automatic way for HR to update the data in the Payroll files, it is not difficult to foresee such inconsistencies arising. Even if Payroll is notified of the changes, it is possible that the data will be entered incorrectly.

Data dependence

As we have already mentioned, the physical structure and storage of the data files and records are defined in the application code. This means that changes to an existing structure are difficult to make. For example, increasing the size of the PropertyForRent address field from 40 to 41 characters sounds like a simple change, but it requires the creation of a one-off program (that is, a program that is run only once and can then be discarded) that converts the PropertyForRent file to the new format. This program has to:

- Open the original PropertyForRent file for reading
- Open a temporary file with the new structure
- Read a record from the original file, convert the data to conform to the new structure, and write it to the temporary file, then repeat this step for all records in the original file
- Delete the original PropertyForRent file
- Rename the temporary file as PropertyForRent

In addition, all programs that access the PropertyForRent file must be modified to conform to the new file structure. There might be many such programs that access the PropertyForRent file. Thus, the programmer needs to identify all the affected programs, modify them, and then retest them. Note that a program does not even have to use the address field to be affected: it only has to use the PropertyForRent file. Clearly, this process could be very time-consuming and subject to error. This characteristic of file-based systems is known as **program–data dependence**.

Incompatible file formats

Because the structure of files is embedded in the application programs, the structures are dependent on the application programming language. For example, the structure of a file generated by a COBOL program may be different from the structure of a file generated by a C program. The direct incompatibility of such files makes them difficult to process jointly.

For example, suppose that the Contracts Department wants to find the names and addresses of all owners whose property is currently rented out. Unfortunately,

Contracts does not hold the details of property owners; only the Sales Department holds these. However, Contracts has the property number (propertyNo), which can be used to find the corresponding property number in the Sales Department's PropertyForRent file. This file holds the owner number (ownerNo), which can be used to find the owner details in the PrivateOwner file. The Contracts Department programs in COBOL and the Sales Department programs in C. Therefore, matching propertyNo fields in the two PropertyForRent files requires that an application developer write software to convert the files to some common format to facilitate processing. Again, this process can be time-consuming and expensive.

Fixed queries/proliferation of application programs

From the end-user's point of view, file-based systems were a great improvement over manual systems. Consequently, the requirement for new or modified queries grew. However, file-based systems are very dependent upon the application developer, who has to write any queries or reports that are required. As a result, two things happened. In some organizations, the type of query or report that could be produced was fixed. There was no facility for asking unplanned (that is, spur-of-the-moment or *ad hoc*) queries either about the data itself or about which types of data were available.

In other organizations, there was a proliferation of files and application programs. Eventually, this reached a point where the DP Department, with its existing resources, could not handle all the work. This put tremendous pressure on the DP staff, resulting in programs that were inadequate or inefficient in meeting the demands of the users, limited documentation, and difficult maintenance. Often, certain types of functionality were omitted:

- There was no provision for security or integrity.
- Recovery, in the event of a hardware or software failure, was limited or nonexistent.
- Access to the files was restricted to one user at a time—there was no provision for shared access by staff in the same department.

In either case, the outcome was unacceptable. Another solution was required.

1.3 Database Approach

All of the previously mentioned limitations of the file-based approach can be attributed to two factors:

(1) The definition of the data is embedded in the application programs, rather than being stored separately and independently.

(2) There is no control over the access and manipulation of data beyond that imposed by the application programs.

To become more effective, a new approach was required. What emerged were the database and the Database Management System (DBMS). In this section, we provide a more formal definition of these terms, and examine the components that we might expect in a DBMS environment.

1.3.1 The Database

Database A shared collection of logically related data and its description, designed to meet the information needs of an organization.

We now examine the definition of a database so that you can understand the concept fully. The database is a single, possibly large repository of data that can be used simultaneously by many departments and users. Instead of disconnected files with redundant data, all data items are integrated with a minimum amount of duplication. The database is no longer owned by one department but is a shared corporate resource. The database holds not only the organization's operational data, but also a description of this data. For this reason, a database is also defined as a *self-describing collection of integrated records*. The description of the data is known as the **system catalog** (or **data dictionary or metadata**—the "data about data"). It is the self-describing nature of a database that provides program–data independence.

The approach taken with database systems, where the definition of data is separated from the application programs, is similar to the approach taken in modern software development, where an internal definition of an object and a separate external definition are provided. The users of an object see only the external definition and are unaware of how the object is defined and how it functions. One advantage of this approach, known as **data abstraction**, is that we can change the internal definition of an object without affecting the users of the object, provided that the external definition remains the same. In the same way, the database approach separates the structure of the data from the application programs and stores it in the database. If new data structures are added or existing structures are modified, then the application programs are unaffected, provided that they do not directly depend upon what has been modified. For example, if we add a new field to a record or create a new file, existing applications are unaffected. However, if we remove a field from a file that an application program uses, then that application program is affected by this change and must be modified accordingly.

Another expression in the definition of a database that we should explain is "logically related." When we analyze the information needs of an organization, we attempt to identify entities, attributes, and relationships. An **entity** is a distinct object (a person, place, thing, concept, or event) in the organization that is to be represented in the database. An **attribute** is a property that describes some aspect of the object that we wish to record, and a **relationship** is an association between entities. For example, Figure 1.6 shows an Entity–Relationship (ER) diagram for part of the *DreamHome* case study. It consists of:

- six entities (the rectangles): Branch, Staff, PropertyForRent, Client, PrivateOwner, and Lease;
- seven relationships (the names adjacent to the lines): *Has, Offers, Oversees, Views, Owns, LeasedBy,* and *Holds*;
- six attributes, one for each entity: branchNo, staffNo, propertyNo, clientNo, ownerNo, and leaseNo.

The database represents the entities, the attributes, and the logical relationships between the entities. In other words, the database holds data that is logically related. We discuss the ER model in detail in Chapters 12 and 13.

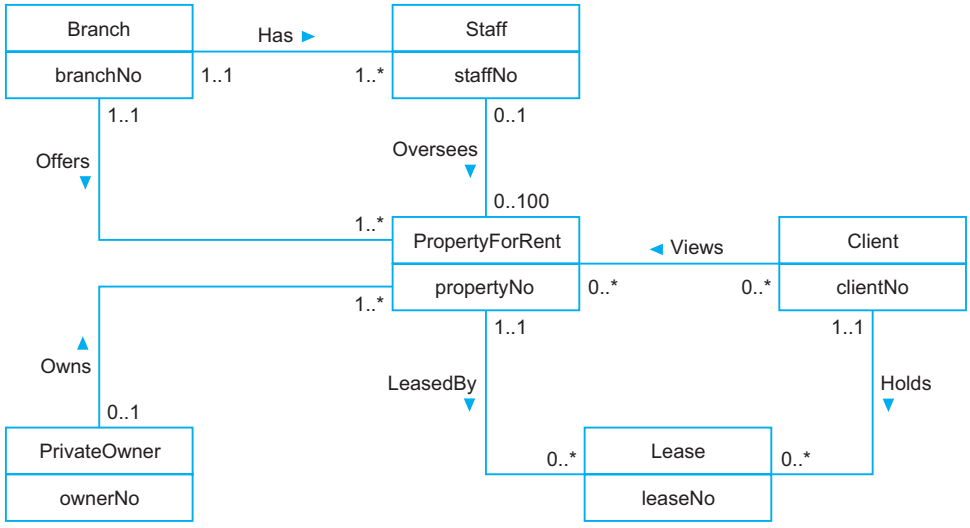


Figure 1.6 Example Entity–Relationship diagram.

1.3.2 The Database Management System (DBMS)

DBMS A software system that enables users to define, create, maintain, and control access to the database.

The DBMS is the software that interacts with the users' application programs and the database. Typically, a DBMS provides the following facilities:

- It allows users to define the database, usually through a **Data Definition Language (DDL)**. The DDL allows users to specify the data types and structures and the constraints on the data to be stored in the database.
- It allows users to insert, update, delete, and retrieve data from the database, usually through a **Data Manipulation Language (DML)**. Having a central repository for all data and data descriptions allows the DML to provide a general inquiry facility to this data, called a **query language**. The provision of a query language alleviates the problems with file-based systems where the user has to work with a fixed set of queries or there is a proliferation of programs, causing major software management problems. The most common query language is the **Structured Query Language** (**SQL**, pronounced "S-Q-L", or sometimes "See-Quel"), which is now both the formal and de facto standard language for relational DBMSs. To emphasize the importance of SQL, we devote Chapters 6–9 and Appendix I to a comprehensive study of this language.
- It provides controlled access to the database. For example, it may provide:
  - a security system, which prevents unauthorized users accessing the database;
  - an integrity system, which maintains the consistency of stored data;
  - a concurrency control system, which allows shared access of the database;
  - a recovery control system, which restores the database to a previous consistent state following a hardware or software failure;
  - a user-accessible catalog, which contains descriptions of the data in the database.

1.3.3 (Database) Application Programs

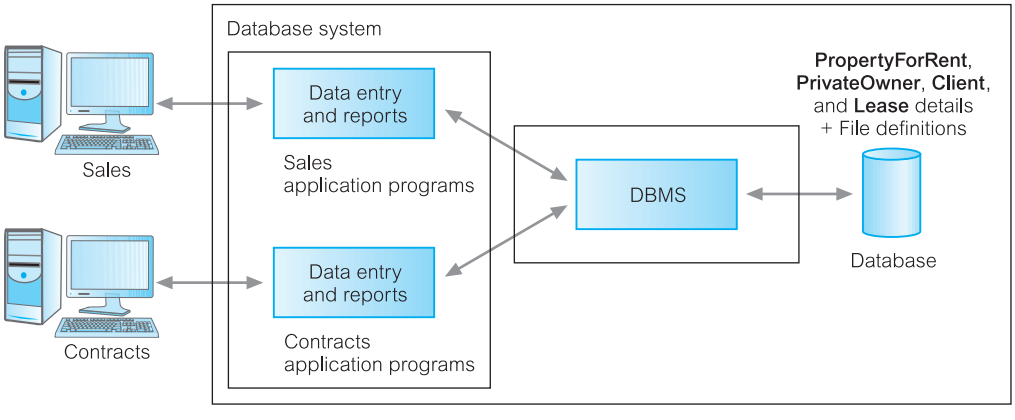
Application Programs — A computer program that interacts with the database by issuing an appropriate request (typically an SQL statement) to the DBMS.

Users interact with the database through a number of **application programs** that are used to create and maintain the database and to generate information. These programs can be conventional batch applications or, more typically nowadays, online applications. The application programs may be written in a programming language or in higher-level fourth-generation language.

The database approach is illustrated in Figure 1.7, based on the file approach of Figure 1.5. It shows the Sales and Contracts Departments using their application programs to access the database through the DBMS. Each set of departmental application programs handles data entry, data maintenance, and the generation of reports. However, as opposed to the file-based approach, the physical structure and storage of the data are now managed by the DBMS.

Views

With this functionality, the DBMS is an extremely powerful and useful tool. However, as the end-users are not too interested in how complex or easy a task is for the system, it could be argued that the DBMS has made things more complex, because they now see more data than they actually need or want. For example, the details that the Contracts Department wants to see for a rental property, as shown in Figure 1.5, have changed in the database approach, shown in Figure 1.7. Now the database also holds the property type, the number of rooms, and the owner details. In recognition of this problem, a DBMS provides another facility known as a **view mechanism**, which allows each user to have his or her own view of the



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)

Figure 1.7 Database processing.

database (a **view** is, in essence, some subset of the database). For example, we could set up a view that allows the Contracts Department to see only the data that they want to see for rental properties.

As well as reducing complexity by letting users see the data in the way they want to see it, views have several other benefits:

- *Views provide a level of security.* Views can be set up to exclude data that some users should not see. For example, we could create a view that allows a branch manager and the Payroll Department to see all staff data, including salary details, and we could create a second view that other staff would use that excludes salary details.
- *Views provide a mechanism to customize the appearance of the database.* For example, the Contracts Department may wish to call the monthly rent field (rent) by the more obvious name, Monthly Rent.
- *A view can present a consistent, unchanging picture of the structure of the database,* even if the underlying database is changed (for example, fields added or removed, relationships changed, files split, restructured, or renamed). If fields are added or removed from a file, and these fields are not required by the view, the view is not affected by this change. Thus, a view helps provide the program–data independence we mentioned previously.

The previous discussion is general and the actual level of functionality offered by a DBMS differs from product to product. For example, a DBMS for a personal computer may not support concurrent shared access, and may provide only limited security, integrity, and recovery control. However, modern, large multi-user DBMS products offer all the functions mentioned and much more. Modern systems are extremely complex pieces of software consisting of millions of lines of code, with documentation comprising many volumes. This complexity is a result of having to provide software that handles requirements of a more general nature. Furthermore, the use of DBMSs nowadays requires a system that provides almost total reliability and 24/7 availability (24 hours a day, 7 days a week), even in the presence of hardware or software failure. The DBMS is continually evolving and expanding to cope with new user requirements. For example, some applications now require the storage of graphic images, video, sound, and so on. To reach this market, the DBMS must change. It is likely that new functionality will always be required, so the functionality of the DBMS will never become static. We discuss the basic functions provided by a DBMS in later chapters.

1.3.4 Components of the DBMS Environment

We can identify five major components in the DBMS environment: hardware, software, data, procedures, and people, as illustrated in Figure 1.8.

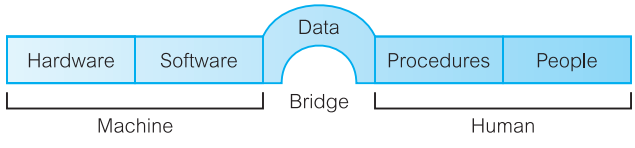


Figure 1.8 The DBMS environment.

Hardware

The DBMS and the applications require hardware to run. The hardware can range from a single personal computer to a single mainframe or a network of computers. The particular hardware depends on the organization's requirements and the DBMS used. Some DBMSs run only on particular hardware or operating systems, while others run on a wide variety of hardware and operating systems. A DBMS requires a minimum amount of main memory and disk space to run, but this minimum configuration may not necessarily give acceptable performance. A simplified hardware configuration for *DreamHome* is illustrated in Figure 1.9. It consists of a network of small servers, with a central server located in London running the **backend** of the DBMS, that is, the part of the DBMS that manages and controls access to the database. It also shows several computers at various locations running the **frontend** of the DBMS, that is, the part of the DBMS that interfaces with the user. This is called a **client–server** architecture: the backend is the server and the frontends are the clients. We discuss this type of architecture in Section 3.1.

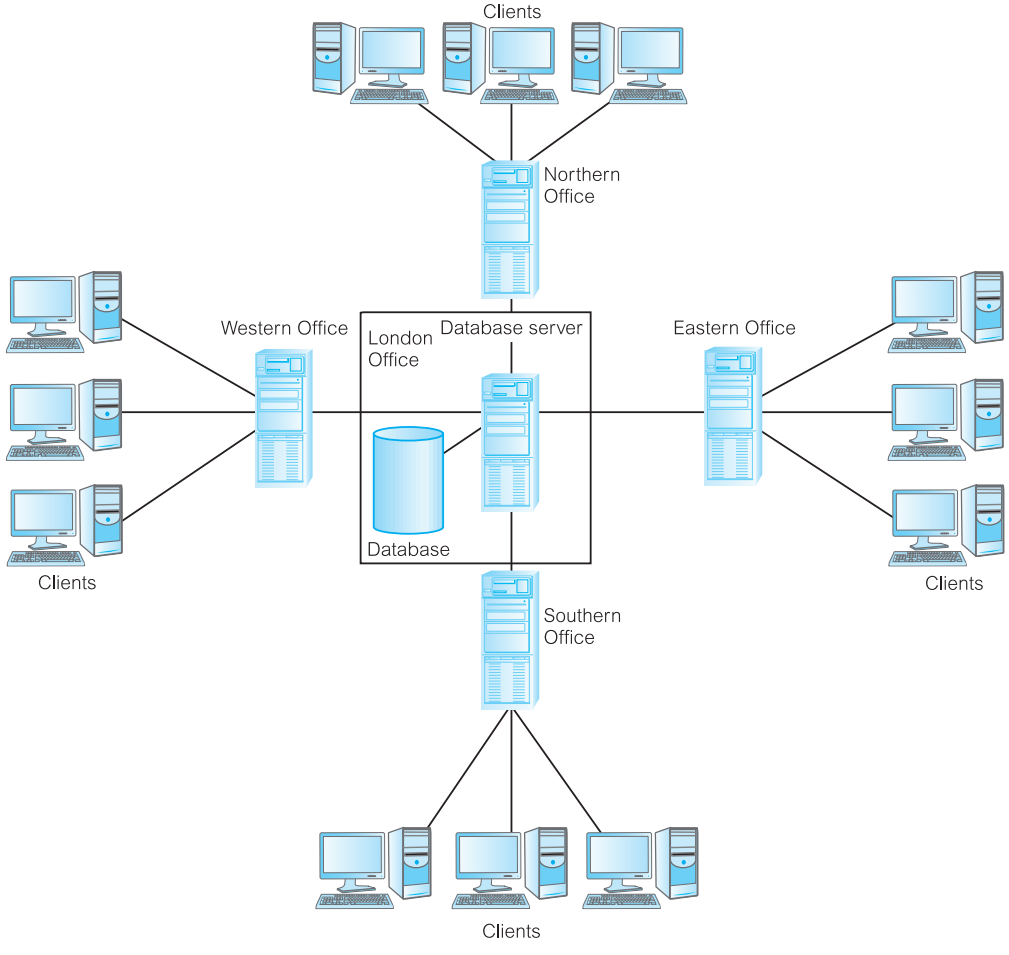


Figure 1.9 *DreamHome* hardware configuration.

Software

The software component comprises the DBMS software itself and the application programs, together with the operating system, including network software if the DBMS is being used over a network. Typically, application programs are written in a third-generation programming language (3GL), such as C, C++, C#, Java, Visual Basic, COBOL, Fortran, Ada, or Pascal, or a fourth-generation language (4GL), such as SQL, embedded in a third-generation language. The target DBMS may have its own fourth-generation tools that allow rapid development of applications through the provision of nonprocedural query languages, reports generators, forms generators, graphics generators, and application generators. The use of fourth-generation tools can improve productivity significantly and produce programs that are easier to maintain. We discuss fourth-generation tools in Section 2.2.3.

Data

Perhaps the most important component of the DBMS environment—certainly from the end-users' point of view—is the data. In Figure 1.8, we observe that the data acts as a bridge between the machine components and the human components. The database contains both the operational data and the metadata, the "data about data." The structure of the database is called the **schema**. In Figure 1.7, the schema consists of four files, or **tables**, namely: PropertyForRent, PrivateOwner, Client, and Lease. The PropertyForRent table has eight fields, or **attributes**, namely: propertyNo, street, city, zipCode, type (the property type), rooms (the number of rooms), rent (the monthly rent), and ownerNo. The ownerNo attribute models the relationship between PropertyForRent and PrivateOwner: that is, an owner *Owns* a property for rent, as depicted in the ER diagram of Figure 1.6. For example, in Figure 1.2 we observe that owner CO46, Joe Keogh, owns property PA14.

The data also incorporates the system catalog, which we discuss in detail in Section 2.4.

Procedures

Procedures refer to the instructions and rules that govern the design and use of the database. The users of the system and the staff who manage the database require documented procedures on how to use or run the system. These may consist of instructions on how to:

- Log on to the DBMS.
- Use a particular DBMS facility or application program.
- Start and stop the DBMS.
- Make backup copies of the database.
- Handle hardware or software failures. This may include procedures on how to identify the failed component, how to fix the failed component (for example, telephone the appropriate hardware engineer), and, following the repair of the fault, how to recover the database.
- Change the structure of a table, reorganize the database across multiple disks, improve performance, or archive data to secondary storage.

People

The final component is the people involved with the system. We discuss this component in Section 1.4.

1.3.5 Database Design: The Paradigm Shift

Until now, we have taken it for granted that there is a structure to the data in the database. For example, we have identified four tables in Figure 1.7: PropertyForRent, PrivateOwner, Client, and Lease. But how did we get this structure? The answer is quite simple: the structure of the database is determined during **database design**. However, carrying out database design can be extremely complex. To produce a system that will satisfy the organization's information needs requires a different approach from that of file-based systems, where the work was driven by the application needs of individual departments. For the database approach to succeed, the organization now has to think of the data first and the application second. This change in approach is sometimes referred to as a *paradigm shift*. For the system to be acceptable to the end-users, the database design activity is crucial. A poorly designed database will generate errors that may lead to bad decisions, which may have serious repercussions for the organization. On the other hand, a well-designed database produces a system that provides the correct information for the decision-making process to succeed in an efficient way.

The objective of this book is to help effect this paradigm shift. We devote several chapters to the presentation of a complete methodology for database design (see Chapters 16–19). It is presented as a series of simple-to-follow steps, with guidelines provided throughout. For example, in the ER diagram of Figure 1.6, we have identified six entities, seven relationships, and six attributes. We provide guidelines to help identify the entities, attributes, and relationships that have to be represented in the database.

Unfortunately, database design methodologies are not very popular. Many organizations and individual designers rely very little on methodologies for conducting the design of databases, and this is commonly considered a major cause of failure in the development of database systems. Owing to the lack of structured approaches to database design, the time or resources required for a database project are typically underestimated, the databases developed are inadequate or inefficient in meeting the demands of applications, documentation is limited, and maintenance is difficult.

1.4 Roles in the Database Environment

In this section, we examine what we listed in the previous section as the fifth component of the DBMS environment: the **people**. We can identify four distinct types of people who participate in the DBMS environment: data and database administrators, database designers, application developers, and end-users.

1.4.1 Data and Database Administrators

The database and the DBMS are corporate resources that must be managed like any other resource. Data and database administration are the roles generally associated with the management and control of a DBMS and its data. The

Data Administrator (**DA**) is responsible for the management of the data resource, including database planning; development and maintenance of standards, policies and procedures; and conceptual/logical database design. The DA consults with and advises senior managers, ensuring that the direction of database development will ultimately support corporate objectives.

The **Database Administrator** (**DBA**) is responsible for the physical realization of the database, including physical database design and implementation, security and integrity control, maintenance of the operational system, and ensuring satisfactory performance of the applications for users. The role of the DBA is more technically oriented than the role of the DA, requiring detailed knowledge of the target DBMS and the system environment. In some organizations there is no distinction between these two roles; in others, the importance of the corporate resources is reflected in the allocation of teams of staff dedicated to each of these roles. We discuss data and database administration in more detail in Section 20.15.

1.4.2 Database Designers

In large database design projects, we can distinguish between two types of designer: logical database designers and physical database designers. The **logical database designer** is concerned with identifying the data (that is, the entities and attributes), the relationships between the data, and the constraints on the data that is to be stored in the database. The logical database designer must have a thorough and complete understanding of the organization's data and any constraints on this data (the constraints are sometimes called **business rules)**. These constraints describe the main characteristics of the data as viewed by the organization. Examples of constraints for *DreamHome* are:

- a member of staff cannot manage more than 100 properties for rent or sale at the same time;
- a member of staff cannot handle the sale or rent of his or her own property;
- a solicitor cannot act for both the buyer and seller of a property.

To be effective, the logical database designer must involve all prospective database users in the development of the data model, and this involvement should begin as early in the process as possible. In this book, we split the work of the logical database designer into two stages:

- conceptual database design, which is independent of implementation details, such as the target DBMS, application programs, programming languages, or any other physical considerations;
- logical database design, which targets a specific data model, such as relational, network, hierarchical, or object-oriented.

The **physical database designer** decides how the logical database design is to be physically realized. This involves:

- mapping the logical database design into a set of tables and integrity constraints;
- selecting specific storage structures and access methods for the data to achieve good performance;
- designing any security measures required on the data.

Many parts of physical database design are highly dependent on the target DBMS, and there may be more than one way of implementing a mechanism. Consequently, the physical database designer must be fully aware of the functionality of the target DBMS and must understand the advantages and disadvantages of each alternative implementation. The physical database designer must be capable of selecting a suitable storage strategy that takes account of usage. Whereas conceptual and logical database design are concerned with the *what*, physical database design is concerned with the *how*. It requires different skills, which are often found in different people. We present a methodology for conceptual database design in Chapter 16, for logical database design in Chapter 17, and for physical database design in Chapters 18 and 19.

1.4.3 Application Developers

Once the database has been implemented, the application programs that provide the required functionality for the end-users must be implemented. This is the responsibility of the **application developers**. Typically, the application developers work from a specification produced by systems analysts. Each program contains statements that request the DBMS to perform some operation on the database, which includes retrieving data, inserting, updating, and deleting data. The programs may be written in a third-generation or fourth-generation programming language, as discussed previously.

1.4.4 End-Users

The end-users are the "clients" of the database, which has been designed and implemented and is being maintained to serve their information needs. End-users can be classified according to the way they use the system:

- **Naïve users** are typically unaware of the DBMS. They access the database through specially written application programs that attempt to make the operations as simple as possible. They invoke database operations by entering simple commands or choosing options from a menu. This means that they do not need to know anything about the database or the DBMS. For example, the checkout assistant at the local supermarket uses a bar code reader to find out the price of the item. However, there is an application program present that reads the bar code, looks up the price of the item in the database, reduces the database field containing the number of such items in stock, and displays the price on the till.
- **Sophisticated users**. At the other end of the spectrum, the sophisticated end-user is familiar with the structure of the database and the facilities offered by the DBMS. Sophisticated end-users may use a high-level query language such as SQL to perform the required operations. Some sophisticated end-users may even write application programs for their own use.

1.5 History of Database Management Systems

We have already seen that the predecessor to the DBMS was the file-based system. However, there was never a time when the database approach began and the file-based system ceased. In fact, the file-based system still exists in specific areas. It has been suggested that the DBMS has its roots in the 1960s Apollo

moon-landing project, which was initiated in response to President Kennedy's objective of landing a man on the moon by the end of that decade. At that time, there was no system available that would be able to handle and manage the vast amounts of information that the project would generate.

As a result, North American Aviation (NAA, now Rockwell International), the prime contractor for the project, developed software known as **GUAM** (for Generalized Update Access Method). GUAM was based on the concept that smaller components come together as parts of larger components, and so on, until the final product is assembled. This structure, which conforms to an upside-down tree, is also known as a **hierarchical structure**. In the mid-1960s, IBM joined NAA to develop GUAM into what is now known as **IMS** (for Information Management System). The reason that IBM restricted IMS to the management of hierarchies of records was to allow the use of serial storage devices; most notably magnetic tape, which was a market requirement at that time. This restriction was subsequently dropped. Although one of the earliest commercial DBMSs, IMS is still the main hierarchical DBMS used by most large mainframe installations.

In the mid-1960s, another significant development was the emergence of **IDS** (or Integrated Data Store) from General Electric. This work was headed by one of the early pioneers of database systems, Charles Bachmann. This development led to a new type of database system known as the **network DBMS**, which had a profound effect on the information systems of that generation. The network database was developed partly to address the need to represent more complex data relationships than could be modeled with hierarchical structures, and partly to impose a database standard. To help establish such standards, the Conference on Data Systems Languages (**CODASYL**), comprising representatives of the U.S. government and the world of business and commerce, formed a List Processing Task Force in 1965, subsequently renamed the **Data Base Task Group (DBTG)** in 1967. The terms of reference for the DBTG were to define standard specifications for an environment that would allow database creation and data manipulation. A draft report was issued in 1969, and the first definitive report was issued in 1971. The DBTG proposal identified three components:

- the network **schema**—the logical organization of the entire database as seen by the DBA—which includes a definition of the database name, the type of each record, and the components of each record type;
- the **subschema**—the part of the database as seen by the user or application program;
- a data management language to define the data characteristics and the data structure, and to manipulate the data.

For standardization, the DBTG specified three distinct languages:

- a schema DDL, which enables the DBA to define the schema;
- a subschema DDL, which allows the application programs to define the parts of the database they require;
- a DML, to manipulate the data.

Although the report was not formally adopted by the American National Standards Institute (ANSI), a number of systems were subsequently developed following the DBTG proposal. These systems are now known as CODASYL or DBTG systems. The CODASYL and hierarchical approaches represented the **first generation** of DBMSs.

We look more closely at these systems on the Web site for this book (see the Preface for the URL). However, these two models have some fundamental disadvantages:

- complex programs have to be written to answer even simple queries based on navigational record-oriented access;
- there is minimal data independence;
- there is no widely accepted theoretical foundation.

In 1970, E. F. Codd of the IBM Research Laboratory produced his highly influential paper on the relational data model ("A relational model of data for large shared data banks," Codd, 1970). This paper was very timely and addressed the disadvantages of the former approaches. Many experimental relational DBMSs were implemented thereafter, with the first commercial products appearing in the late 1970s and early 1980s. Of particular note is the System R project at IBM's San José Research Laboratory in California, which was developed during the late 1970s (Astrahan et al., 1976). This project was designed to prove the practicality of the relational model by providing an implementation of its data structures and operations, and led to two major developments:

- the development of a structured query language called SQL, which has since become the standard language for relational DBMSs;
- the production of various commercial relational DBMS products during the 1980s, for example DB2 and SQL/DS from IBM and Oracle from Oracle Corporation.

Now there are several hundred relational DBMSs for both mainframe and PC environments, though many are stretching the definition of the relational model. Other examples of multi-user relational DBMSs are MySQL from Oracle Corporation, Ingres from Actian Corporation, SQL Server from Microsoft, and Informix from IBM. Examples of PC-based relational DBMSs are Office Access and Visual FoxPro from Microsoft, Interbase from Embarcadero Technologies, and R:Base from R:Base Technologies. Relational DBMSs are referred to as **second-generation** DBMSs. We discuss the relational data model in Chapter 4.

The relational model is not without its failings; in particular, its limited modeling capabilities. There has been much research since then attempting to address this problem. In 1976, Chen presented the Entity–Relationship model, which is now a widely accepted technique for database design and the basis for the methodology presented in Chapters 16 and 17 of this book. In 1979, Codd himself attempted to address some of the failings in his original work with an extended version of the relational model called RM/T (1979) and subsequently RM/V2 (1990). The attempts to provide a data model that represents the "real world" more closely have been loosely classified as **semantic data modeling**.

In response to the increasing complexity of database applications, two "new" systems have emerged: the **object-oriented DBMS (OODBMS)** and the **object-relational DBMS (ORDBMS)**. However, unlike previous models, the actual composition of these models is not clear. This evolution represents **third-generation** DBMSs, which we discuss in Chapters 9 and 27–28.

The 1990s also saw the rise of the Internet, the three-tier client-server architecture, and the demand to allow corporate databases to be integrated with Web applications. The late 1990s saw the development of XML (eXtensible Markup Language), which has had a profound effect on many aspects of IT, including

database integration, graphical interfaces, embedded systems, distributed systems, and database systems. We will discuss Web–database integration and XML in Chapters 29–30.

Specialized DBMSs have also been created, such as **data warehouses**, which can store data drawn from several data sources, possibly maintained by different operating units of an organization. Such systems provide comprehensive data analysis facilities to allow strategic decisions to be made based on, for example, historical trends. All the major DBMS vendors provide data warehousing solutions. We discuss data warehouses in Chapters 31–32. Another example is the **enterprise resource planning (ERP)** system, an application layer built on top of a DBMS that integrates all the business functions of an organization, such as manufacturing, sales, finance, marketing, shipping, invoicing, and human resources. Popular ERP systems are SAP R/3 from SAP and PeopleSoft from Oracle. Figure 1.10 provides a summary of historical development of database systems.

TIMEFRAME	DEVELOPMENT	COMMENTS
1960s (onwards)	File-based systems	Precursor to the database system. Decentralized approach: each department stored and controlled its own data.
Mid-1960s	Hierarchical and network data models	Represents first-generation DBMSs. Main hierarchical system is IMS from IBM and the main network system is IDMS/R from Computer Associates. Lacked data independence and required complex programs to be developed to process the data.
1970	Relational model proposed	Publication of E. F. Codd's seminal paper "A relational model of data for large shared data banks," which addresses the weaknesses of first-generation systems.
1970s	Prototype RDBMSs developed	During this period, two main prototypes emerged: the Ingres project at the University of California at Berkeley (started in 1970) and the System R project at IBM's San José Research Laboratory in California (started in 1974), which led to the development of SQL.
1976	ER model proposed	Publication of Chen's paper "The Entity-Relationship model—Toward a unified view of data." ER modeling becomes a significant component in methodologies for database design.
1979	Commercial RDBMSs appear	Commercial RDBMSs like Oracle, Ingres, and DB2 appear. These represent the second generation of DBMSs.
1987	ISO SQL standard	SQL is standardized by the ISO (International Standards Organization). There are subsequent releases of the standard in 1989, 1992 (SQL2), 1999 (SQL:1999), 2003 (SQL:2003), 2008 (SQL:2008), and 2011 (SQL:2011).
1990s	OODBMS and ORDBMSs appear	This period initially sees the emergence of OODBMSs and later ORDBMSs (Oracle 8, with object features released in 1997).
1990s	Data warehousing systems appear	This period also see releases from the major DBMS vendors of data warehousing systems and thereafter data mining products.
Mid-1990s	Web–database integration	The first Internet database applications appear. DBMS vendors and third-party vendors recognize the significance of the Internet and support web–database integration.
1998	XML	XML 1.0 ratified by the W3C. XML becomes integrated with DBMS products and native XML databases are developed.

Figure 1.10 Historical development of database systems.

1.6 Advantages and Disadvantages of DBMSs

The database management system has promising potential advantages. Unfortunately, there are also disadvantages. In this section, we examine these advantages and disadvantages.

Advantages

The advantages of database management systems are listed in Table 1.2.

Control of data redundancy As we discussed in Section 1.2, traditional file-based systems waste space by storing the same information in more than one file. For example, in Figure 1.5, we stored similar data for properties for rent and clients in both the Sales and Contracts Departments. In contrast, the database approach attempts to eliminate the redundancy by integrating the files so that multiple copies of the same data are not stored. However, the database approach does not eliminate redundancy entirely, but controls the amount of redundancy inherent in the database. Sometimes it is necessary to duplicate key data items to model relationships; at other times, it is desirable to duplicate some data items to improve performance. The reasons for controlled duplication will become clearer as you read the next few chapters.

Data consistency By eliminating or controlling redundancy, we reduce the risk of inconsistencies occurring. If a data item is stored only once in the database, any update to its value has to be performed only once and the new value is available immediately to all users. If a data item is stored more than once and the system is aware of this, the system can ensure that all copies of the item are kept consistent. Unfortunately, many of today's DBMSs do not automatically ensure this type of consistency.

More information from the same amount of data With the integration of the operational data, it may be possible for the organization to derive additional information from the same data. For example, in the file-based system illustrated in Figure 1.5, the Contracts Department does not know who owns a leased property. Similarly, the Sales Department has no knowledge of lease details. When we

TABLE 1.2 Advantages of DBMSs.

Control of data redundancy	Economy of scale
Data consistency	Balance of conflicting requirements
More information from the same amount of data	Improved data accessibility and responsiveness
Sharing of data	Increased productivity
Improved data integrity	Improved maintenance through data independence
Improved security	Increased concurrency
Enforcement of standards	Improved backup and recovery services